Daily Combinatorics Drill #7

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Section	Topic	Questions	Target time
A	Rapid Recognition	10	2 min 30 sec
B	Manipulation Drills	10	8 min
C	Substitution & Structure	8	10 min
D	Challenge	5	15 min

New toolkit today: Recurrences & Fibonacci Counting · Catalan Numbers · Bijections · Invariants & Monovariants.

Section A Rapid Recognition (≤15 s each)

1. Climb 4 stairs, steps of 1 or 2?

Ans: 5

2. Climb 5 stairs?

Ans: 8

3. Binary strings of length 4, no two adjacent 1s?

Ans: 8

4. Domino tilings of 2×3?

Ans: 3

5. Domino tilings of 2×4?

Ans: 5

6. Regions from 3 lines in general position?

Ans: 7

7. Correctly matched arrangements of 3 bracket pairs?

Ans: 5

8. $a_1 = 1$, $a_{n+1} = 2a_n + 1$: find a_4.

Ans: 15

9. Write F_8.

Ans: 21

10. True or false: $F_1 + \cdots + F_8 = 54$.

Ans: True

Section B Manipulation Drills ($\leq$50 s each)

1. Climb 10 stairs, steps of 1 or 2?

Ans: 89

2. Binary strings of length 8, no two adjacent 1s?

Ans: 55

3. Domino tilings of 2×10?

Ans: 89

4. Regions from 10 lines in general position?

Ans: 56

5. $a_1 = 2$, $a_n = 3a_{n-1} - 1$: find a_4.

Ans: 41

6. Climb 6 stairs, steps of 1, 2, or 3?

Ans: 24

7. Regions of a disc from 5 points, all chords?

Ans: 16

8. Snaps to break a 4×6 bar into 24 squares (no stacking)?

Ans: 23

9. Least moves, Towers of Hanoi, 6 disks?

Ans: 63

10. Use $F_1 + \cdots + F_n = F_{n+2} - 1$ to evaluate $F_4 + \cdots + F_{12}$.

Ans: 372

Section C Substitution & Structure ($\leq$90 s each)

1. Climb 8 stairs, steps 1 or 2, no two 2-steps in a row?

Ans: 19

2. Tile a 2×6 strip with dominoes and 2×2 squares?

Ans: 43

3. Lattice paths $(0,0) \to (4,4)$, unit steps, never above $y = x$?

Ans: $C_4 = 14$

4. Regions of a disc from 6 points, all chords? *(not 32)*

Ans: 31

5. Ordered sums of 8 into odd parts?

Ans: 21

6. Length-6 strings over $\{0, 1, 2\}$ with no two adjacent 0s?

Ans: 448

7. Domino tilings of a 3×4 grid?

Ans: 11

8. Binary strings of length 10 with no three consecutive 1s?

Ans: 504

Section D Challenge (BMO1 difficulty)

1. Permutations with $|p(i) - i| \leq 1$: prove $a_n = a_{n-1} + a_{n-2}$; find a_6.

Ans: $a_6 = 13$

2. Erase two numbers, write their difference, repeat: prove the last number is odd.

Ans: Proof below.

3. Prove a convex hexagon has 14 triangulations.

Ans: $C_4 = 14$

4. Prove any $2^n \times 2^n$ board minus one square is L-tromino tileable.

Ans: Proof below (induction).

5. Binary strings of length 5 with no two consecutive 1s are listed (there are 13). Consider a string and statements: (A) it starts with 1, (B) it contains exactly two 1s, (C) it ends with 0. Exactly one of the three statements is true. How many strings satisfy this? *(after TMUA Practice Paper 2 Q4)*

Ans: 7